

Emma Allen

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I am a DPhil student in the Intelligent Earth Doctoral Programme at the University of Oxford. I aim to build a career in climate physics and machine learning. I am still deciding on a firm research direction, although I am currently interested in hybrid climate modelling following my masters thesis on NeuralGCM.

Education

University of Oxford 2026 to 2030

Intelligent Earth Centre for Doctoral Training, DPhil

- Courses across machine learning, research computing and climate physics

University of Cambridge 2025 to 2026

Data Intensive Science, MPhil

- Courses across machine learning, Bayesian statistics, high performance computing, reinforcement learning, weather and climate, and high energy physics.
- Awarded the Murray Edwards Scholarship for Women in Data Intensive Science

University of Edinburgh 2021 to 2025

Computational Physics, BSc with Honours

- Awarded a pre-honours certificate of achievement for excellent grades in the first and second years.
- Awarded the Margaret Campbell Scott Scholarship for excellent entrance qualifications.

University of California, Los Angeles, UCLA 2023 to 2024

Academic exchange

- Completed courses in probability (A), machine learning (A), and differential equations (A)
- Attended the Conference for Undergraduate Women and Non-binary Physicists in San Diego.

Research Experience

MPhil Thesis January to July 2026

University of Cambridge

- Evaluated NeuralGCM as a hybrid physics and machine learning model for weather and climate prediction.
- Reproduced forecast accuracy and tropical cyclone results from the original NeuralGCM paper.
- Extended the evaluation with additional tropical cyclone case studies focusing on track curvature, intensity, re-intensification, and Fujiwhara interaction.
- Found that NeuralGCM partly captured cyclone tracks but consistently underestimated intensity. Deterministic models successfully captured curvature and Fujiwhara interaction, but missed the re-intensification.
- Supervisors: Professor Sebastian Schemm, Dr Simon Driscoll, and Pritthijit Nath.
- *Tools utilised* Python, LaTeX, Xarray and the JASMIN HPC system.

Summer Research Student June to August 2025

University of Edinburgh

- Investigated nonparametric reconstruction of parton distribution functions and the effect of Gaussian assumptions in NNPDF4.0 on physical observable predictions.
- Assessed one dimensional replicas using excess kurtosis and reconstructed distributions in one and two dimensions using kernel density estimation.
- Developed a method to construct a 450 dimensional global correlation matrix from two dimensional covariance estimates, obtaining strong agreement with empirical results.
- Awarded the School of Physics Summer Research Scholarship, worth £2,580.
- Supervisor: Professor Luigi Del Debbio.
- *Tools utilised* Python and LaTeX.

Senior Honours Project September to December 2024

University of Edinburgh

- Modelled a curved bacterium as a polymer in constant shear flow using molecular dynamics within

multiparticle collision dynamics.

- Studied rotational behaviour as curvature varied from 0 to 2π , finding that noise influenced the dynamics more strongly than curvature and produced deviations from theoretical predictions.
- Supervisors: Dr Tyler N. Shendruk and Dr Gavin Melaugh.
- *Tools utilised* Python, Bash, and LaTeX.

Undergraduate Researcher

January to August 2024

University of California, Los Angeles

- Analysed surface temperature variability from 1980 to 2020 using de-trending, basic statistics, and Empirical Orthogonal Function analysis.
- Used singular value decomposition to identify dominant variability patterns in North America and Eurasia.
- Supervisor: Professor Gang Chen.
- *Tools utilised* Python, NumPy, scikit learn, PyTorch, and LaTeX.

Industry Experience

Research Intern

June to August 2024

Applied Science and Technology Research Institute, Hong Kong

- Investigated compression of model weights and key value cache data for large language models.
- Implemented compression methods in C++ and used entropy based techniques to achieve a compression ratio of 92.4%, close to the theoretical limit of 92.1%.
- Documented the methods and results in a technical report and presentation.
- *Tools utilised* C++, Python, and LaTeX.

Spring Intern

April 2023

Optiver, Amsterdam

- Developed a dual listing trading algorithm in Python as part of a team that placed fourth in the Optibook challenge.

Safety Case Engineer

November 2020 to

EDF, Gloucestershire

September 2021

- Authored safety case documentation for nuclear power station defuelling and reviewed fleet chemistry standards.
- Consolidated research on the effects of climate change across the nuclear fleet and presented recommendations to colleagues and members of the Office for Nuclear Regulation.
- Maintained an internal information website to consolidate organisational and operational information.
- Achieved the Industrial Cadets Platinum Leaders Award from the Engineering Development Trust.

Mathematics and Physics Tutor

2018 to 2023

Various locations

- Designed and delivered individual lessons, adapting explanations and problem solving strategies to each student.

Projects

Astronomy Project

2023

- Led the Python team and developed object oriented software to reproduce the analysis supporting Hubble's conclusion on the expansion of the Universe. The poster and project summary available [here](#).

Students as Change Agents

2022

- Produced a video and report outlining short, medium, and long term solutions for saving vacant buildings in Scotland.
- Presented the recommendations to Grand Bequest and completed the Edinburgh Award.

Additional Experience and Awards

Cambridge Journal of Climate Research Associate Editor, 2026 Served as an Associate Editor for both 2026 issues of this interdisciplinary journal, which publishes climate research spanning science, engineering, law, policy, politics and economics.

PADI Divemaster, 2025 Qualified scuba diver.

University society positions, 2021 to 2026 Research Team Member for the Cambridge Climate Society in 2026, contributing to its newsletter and helping to deliver CamCOP and research presentation evenings for early career researchers. Head of socials for Hoppers, a society for gender minorities in computing, and head of media and marketing for the Hot Chocolate Society from 2021 to 2023.

Grade Eight Piano, 2023 Achieved after playing piano for five years.

Engineering Development Trust Ambassador, 2021 to 2024 Delivered panel sessions about industrial placements.

Newnham College Essay Competition, 2019 Awarded first place for an essay on the future of autonomous vehicles.

Technologies

Programming languages Python, C++, C, Java, Haskell, HTML, CSS, and Bash.

Scientific computing PyTorch, TensorFlow, scikit learn, NumPy, xarray, JAX, high performance computing, machine learning, statistical analysis, numerical modelling, and scientific visualisation.